GRIDEDGE ANALYTICS · PREPARED FOR SCUDERIA FERRARI HP

Ground Effect Era

Tire Strategy & Pace Management Analysis

2022 – 2025 · F1 Regulations Cycle · Confidential

4 Seasons	~90 Races	3 Analyses
Complete regulation cycle	All dry-tire race events	Degradation · Pit strategy · Pace

01 Business Task

Defining the analytical question

Why Ferrari?

Scuderia Ferrari is the ideal subject for a tire strategy analysis precisely because the gap between their car's raw performance and their race results across the ground effect era is not explained by pace. Ferrari has produced competitive machinery in each season from 2022 onward — Charles Leclerc took pole positions, set fastest laps, and led races that Ferrari ultimately did not win. The shortfall has been consistently attributed to strategic execution: pit timing calls that were made too late, compound selections that did not account for degradation trajectory, and a reactive rather than proactive pit wall culture.

This analysis was built to quantify what a data-driven strategy approach would look like for a team in Ferrari's position — fast enough to win, but not yet winning consistently enough to match their underlying performance.

Statement of Task

GridEdge Analytics was engaged to answer the following strategic question for Scuderia Ferrari HP ahead of the 2025 season:

How do race-winning drivers and mid-field drivers use tire strategy and pace management differently across the ground effect regulation era (2022–2025), and what specific adjustments should Ferrari's strategy team make to close the gap between raw pace and race results?

Supporting Questions

- **Q1** How do tire degradation rates differ by compound across high-deg and low-deg circuits — and how does this change the viable pit window?
- **Q2** Which compound transitions are most associated with podium finishes across the era, and how does this vary by circuit degradation type?
- **Q3** Has tire management difficulty changed as teams developed their ground effect cars — and if so, does Ferrari's strategic approach need to adapt to a lower-degradation environment?
- **Q4** Which drivers most consistently outperformed their raw pace through strategy, and what behavioral patterns distinguish them?

Key Stakeholder

Head of Race Strategy, Scuderia Ferrari HP. This deliverable feeds directly into pre-season strategic planning, circuit-by-circuit compound allocation decisions, and driver coaching priorities for tire management in Stints 2 and 3.

02 The Ground Effect Era — Technical Context

Why the 2022 regulations changed everything about strategy

What Is Ground Effect?

Ground effect is an aerodynamic principle where a car generates the majority of its downforce from airflow beneath the floor rather than from wings and bodywork above the car. Air traveling under a specially shaped underfloor accelerates and creates a low-pressure region that effectively pulls the car toward the track surface. The faster the car goes, the stronger the suction — meaning downforce scales with speed in a fundamentally different way than wing-based aerodynamics.

F1 banned ground effect floors in 1983 after a series of accidents exposed how suddenly the cars lost downforce when the underfloor seal was disrupted. For nearly four decades, teams relied primarily on upper-body aerodynamics — front wings, rear wings, bargeboards, and diffusers — to generate grip. Ferrari competed through this entire era and built deep institutional knowledge around wing-led development. The 2022 reset upended that knowledge base across the paddock.

The 2022 Regulation Reset

The FIA introduced a sweeping technical overhaul for 2022 with two primary goals: reduce aerodynamic turbulence so cars could race more closely, and contain development costs. The headline change was the reintroduction of ground effect via Venturi tunnels — curved channels running through the underfloor that accelerate airflow to generate downforce. Front wings were simplified. Bargeboards were banned. The floor became the primary aerodynamic surface of the car.

	Pre-2022 Hybrid Era	2022–2025 Ground Effect Era
Primary downforce source	Front wing, bargeboards, rear diffuser	Venturi floor tunnels
Wing complexity	Very high — multi-element front wing	Simplified — reduced elements
Bargeboards	Permitted and extensively developed	Banned entirely
Minimum car weight	752 kg	798 kg (+46 kg)
Wheel diameter	13-inch	18-inch
Tire sidewall profile	Tall / high flex	Low profile / stiff
Ferrari's relative position	Competitive but not dominant	Front row pace, strategic gaps

Porpoising and Its Effect on Ferrari's 2022 Season

The earliest ground effect cars suffered from porpoising — a violent oscillation where the car repeatedly lost and regained floor seal at high speed. When the underfloor sealed completely, downforce spiked and pushed the car down. This compressed the suspension until the floor touched the track, disrupting airflow, collapsing downforce, and bouncing the car back up. The cycle repeated at frequencies up to 5 times per second.

Ferrari entered 2022 with one of the stronger car concepts and was immediately competitive. However, like all teams, they faced porpoising-driven setup constraints that forced compromises in ride height and suspension stiffness. Running higher to reduce porpoising cost downforce and changed how the tires were loaded — transferring more mechanical stress to the contact patch. This elevated degradation in the early part of the era and contributed to variability in race results that would not have appeared in clean testing data.

The practical consequence for Ferrari was that 2022 race setup required balancing car performance against ride stability in a way that was not always consistent with optimal tire strategy. A car that porpoises needs a stiffer setup; a stiffer setup transfers more load to the tires; more tire load produces higher degradation rates on abrasive circuits.

18-Inch Wheels and How They Changed Tire Strategy

The switch from 13-inch to 18-inch wheels with Pirelli's new low-profile construction was among the most strategically significant changes of the regulation reset. The previous 13-inch wheel used a tall, flexible sidewall that acted as secondary suspension — absorbing bumps and mechanical load through flex, which also generated frictional heat and contributed to thermal degradation.

The new 18-inch tire has a much stiffer sidewall with minimal flex. This changed how tires generate and manage heat in two important ways that directly affect the strategic calculations Ferrari must make:

- Thermal degradation became less dominant — the new tires are more thermally stable and less likely to overheat through sidewall flex alone. The degradation mechanism shifted toward mechanical grain and blistering driven by surface interaction.
- Warming up takes longer — the rigid construction is slower to reach optimal operating temperature, which changed the value of fresh tires on out-laps and affected the pace advantage available immediately after a pit stop.
- Degradation became more driver-sensitive — without the thermal buffer of flexible sidewalls, mechanical inputs from the driver (lateral slip, brake bias, throttle application through corners) translate more directly into surface wear. This amplifies the difference between drivers who protect tires and those who abuse them.

How Ground Effect Changed Race Strategy

1. Following Cars More Closely Became Possible — But Changed the Undercut Calculus

Ground effect cars retain more downforce when following closely. Real-world testing suggested cars retain approximately 70–75% of their downforce at one car length, compared to roughly 55% in the hybrid era. Drivers can apply pressure for longer before their tires overheat from aerodynamic instability. This sounds like it would make overtaking easier — but it actually made the undercut harder to execute because fresh tires provide relatively less pace advantage over a car that is still maintaining grip while being followed closely.

For Ferrari, whose strategy often relied on the undercut as a primary overtaking mechanism, this was a meaningful change. The timing window for a successful undercut narrowed. Executing it too late gave rivals time to react. Executing it too early sacrificed track position unnecessarily. The margin for error compressed.

2. High-Deg Circuits Became More Strategically Critical

The stiff low-profile tires interact more aggressively with abrasive circuit surfaces. At venues like Bahrain, Spain, and Hungary — high lateral loads, abrasive tarmac, sustained corner sequences — the mechanical interaction between tire and track creates granulation and blistering that accelerates in ways that are less predictable and more driver-sensitive than thermal degradation alone. This is where Ferrari's strategic decisions carry the most weight, because degradation variance between drivers is highest and the penalty for a wrong compound choice or a late pit call is largest.

3. Car Development Changed the Strategic Environment Year-by-Year

Unlike previous eras where the regulatory framework was stable, the ground effect era saw significant floor development as teams optimized their Venturi tunnels. A more efficient floor generates more downforce at lower drag, which changes how tires are loaded. More efficient downforce means less mechanical slip and lower tire stress at a given cornering speed. This means the strategic environment has not been static across 2022–2025.

For Ferrari's strategy team, this creates a compounding challenge: the compound selection and pit window models that worked in 2022 may have become suboptimal by 2024 as the field's degradation rates fell. A strategy built on 2022 assumptions applied in a 2024 race could be wrong not because of bad execution but because the underlying tire behavior changed.

03 Data Sources

How the dataset was assembled

FastF1 API

All data was pulled directly from the FastF1 Python library, which provides access to F1's official timing feed data across all sessions from 2018 onward. For this analysis we use the full 2022–2025 race calendar.

Parameter	Value
Source	FastF1 v3.x Python library — F1 official timing feed
Coverage	2022, 2023, 2024, 2025 race sessions (R only — no qualifying or practice)
Compounds analyzed	SOFT, MEDIUM, HARD — dry tires only
Lap exclusions	In/out laps, wet compound laps, laps >110% of race fastest lap
Pit stop detection	Derived from PitInTime column in session lap data
Normalization	Z-score per race — enables cross-circuit and cross-season comparison
Position groups	Podium = P1–P3, Top 10 = P4–P10, Back = P11+
Minimum sample	5 laps for degradation regression, 10 race starts for strategy delta ranking

Key Fields

- LapTime — converted from Python timedelta to float seconds; basis for all pace calculations
- TyreLife — laps completed on the current set; x-variable in degradation regression
- Compound — tire type per lap; filtered to dry compounds for this analysis
- PitInTime / PitOutTime — identifies pit stop laps and marks in/out laps for removal
- Stint — stint number per driver per race; used to reconstruct compound sequences
- Position / GridPosition — from session results object; used for position groups and strategy delta

Limitations

- Pit stop duration is not consistently populated in FastF1 across all sessions — stop duration analysis is not included

- In-race position tracking (lap-by-lap position changes) is not used in this version — strategy delta is computed from final finish position vs. pace rank
- 2025 data is partial — rounds completed before the analysis run date are included; upcoming rounds are skipped
- Wet race sessions are fully excluded — intermediate and wet tire strategy follows separate logic not addressed here

04 Data Cleaning Documentation

Every step taken and why

Cleaning Pipeline

Each step below is logged with a before/after row count in the Python script console output, making the cleaning process fully auditable and reproducible.

Step	Action	Reason	Impact
1	Null lap time drop	LapTime is null on formation laps, laps where timing resets behind a safety car, and some sessions where the F1 timing system did not record the lap correctly.	< 1% of rows
2	In lap and out lap removal	PitInTime not null means the driver entered pits at the end of this lap. PitOutTime not null means they exited pits this lap. Both are structurally slow and not representative of racing pace.	~5–8% of rows
3	Wet compound filter	INTERMEDIATE and WET laps follow completely different strategic logic and cannot be compared to dry strategy. Removed to keep the degradation and pace analysis internally consistent.	Varies — wet sessions excluded entirely
4	110% fastest lap filter	Any lap more than 10% slower than the fastest clean lap in that race is classified as a safety car, VSC, or incident lap. Retaining these would depress average pace and artificially lower degradation slopes.	~3–5% of rows
5	DNF/DSQ removal	Drivers who did not finish have no valid finish position and cannot be assigned to a position group. Excluding them prevents null values from propagating through position-based analyses.	~2–3 entries per race
6	Race-level z-score normalization	Lap times differ by 40+ seconds between circuits. Normalizing within each individual race (rather than by circuit name) enables cross-season pace comparison without circuit or year bias. A normalized value of 0 means exactly average pace for that race.	Column added — no rows removed

05 Analysis Summary

Key findings across all three analytical threads

Thread 1 — Tire Degradation Rates

Degradation was measured using ordinary least squares linear regression of lap time against lap-in-stint number. The slope of the regression line represents seconds added per additional lap on a given tire set.

Finding 1.1 — Compound Hierarchy Is Consistent

Across all four seasons and all position groups, SOFT tires degrade fastest, MEDIUM in the middle, and HARD slowest. The important finding is the magnitude of the difference by circuit type. On high-deg circuits, SOFT degradation runs at roughly 3x the rate of HARD. On low-deg circuits the difference nearly disappears — SOFT is sometimes marginally better than HARD because the fresh-tire grip advantage over a minimal wear rate makes extending a SOFT stint viable.

Finding 1.2 — Ferrari's Critical Circuits Are Disproportionately High-Deg

Of Ferrari's historically strong circuits — Bahrain, Spain, Monza, Hungary, Silverstone — the majority fall into the high-deg category. This is significant because Ferrari's strategic errors have been most visible at exactly these venues. The data suggests that the compound selection and pit timing decisions at high-deg circuits carry a disproportionate result impact, and any strategic model Ferrari uses needs to be calibrated most precisely for this circuit type.

Finding 1.3 — Degradation Rates Declined Across the Era

MEDIUM tire degradation rates show a consistent decline from 2022 to 2024. This aligns with the technical development story: more efficient Venturi floors generate downforce with less mechanical slip, reducing shear forces on the tire contact patch. The practical implication is that race strategies requiring two stops in 2022 became viable one-stop strategies on the same circuits by 2024. A strategy team still calibrating on 2022 degradation data in a 2024 race is working from an outdated model.

Thread 2 — Pit Stop Timing and Compound Transitions

Finding 2.1 — Podium Finishers Are Overrepresented in Early Pit Windows

When pit stops are categorized as early (before 33% race distance), mid (33–66%), or late (after 66%), podium finishers show a higher proportion of early stops compared to mid-field and backmarker groups. This is consistent with Ferrari's known challenge: their pit call culture has historically been reactive, responding to tire data rather than pre-empting

rivals. The data across the era shows the teams finishing on podiums are more frequently initiating the strategic play rather than responding to it.

Important caveat: this is correlational. The fastest cars are also most likely to be in a position where an early pit stop is viable — they can pit earlier because they have the pace to rejoin in clean air without losing net position. But even controlling for car pace, early stop timing correlates with better outcomes across mid-field teams too, which is the actionable signal for Ferrari.

Finding 2.2 — SOFT to HARD Is the Most Effective Opening Strategy

Across all four seasons, the SOFT to HARD compound transition shows the highest association with podium finishes. The strategic logic is clear: an aggressive SOFT opening stint builds an early pace gap, then a long HARD stint closes out the race with minimal degradation cost. This strategy is most viable on low-to-medium degradation circuits where SOFT can complete 12–18 laps before its degradation cost erases the pace benefit. On high-deg circuits the SOFT window shrinks to 10 laps or fewer, which compresses the timing margin available to execute the transition successfully.

Finding 2.3 — Pit Stop Count Differences Between Groups Are Small

The average number of pit stops per race across position groups differs by less than 0.2 across the era. Winning is not about choosing a different number of stops — it is about the timing and compound selection of those stops. A one-stop strategy can produce a podium or a tenth-place finish depending entirely on when it happens and what compounds are used. This is directly relevant to Ferrari, where the decision to pit or stay out has repeatedly been made a lap or two later than the optimal window.

Thread 3 — Pace Consistency and Strategy Delta

Finding 3.1 — Top-Tier Drivers Show Lower Lap-Time Variance

Using the standard deviation of race-normalized lap times as a consistency metric, top-tier drivers show lower variance than mid-field and backmarker drivers. This effect persists even within the same finishing group — drivers in the top 10 who score consistently tend to have lower lap-time variance than top-10 drivers who score inconsistently. For Ferrari, where both Leclerc and Sainz/Hamilton have been fast in qualifying but inconsistent through race stints, lap-time variance in Stints 2 and 3 is a meaningful development target.

Finding 3.2 — Strategy Delta Identifies Where Ferrari Left Points

The strategy delta metric — pace rank minus finish position — quantifies how much each driver gains or loses relative to what their raw pace should have produced. Drivers with consistently negative strategy deltas are losing positions that their car's pace should have held. Across the ground effect era, the Ferrari drivers show a mixed picture: fast average pace, but strategy deltas that are less positive (or negative) in races that contained reactive pit decisions versus races where Ferrari took the initiative.

Finding 3.3 — The Strategic Value Pool Is Measurable

The scatter of raw pace rank versus finish position across the era shows a wide cloud of points around the diagonal. The width of that cloud is the total strategic value available — the gap between what the car's pace predicts and what the final result is. In the ground effect era this cloud is wider than it was in the hybrid era, consistent with the regulation intent of making strategy more influential on outcomes. For Ferrari, this represents quantifiable championship points that are sitting in the gap between their car's performance and their race results.

06 Recommendations

Three actionable outputs for Scuderia Ferrari HP

RECOMMENDATION 01 **Pit Timing — Commit to the Early Window on High-Deg Circuits**

Ferrari's documented strategic weakness across the ground effect era has been a tendency to pit one to two laps later than the optimal window, particularly on high-deg circuits. The era-wide data shows podium finishers are disproportionately represented in early pit stop windows (before 33% race distance) at high-deg venues. On these circuits, tire degradation is aggressive enough that the lap-time loss from a competitor on worn tires is visible in the data as early as lap 10–12 of a stint. The recommended strategic posture is to define a pre-race pit window for each circuit and hold to it proactively rather than waiting for tire data to confirm the call. The cost of pitting one lap early is consistently lower than the cost of pitting one lap late when a rival reacts to your stop.

Priority circuits: Bahrain, Spain, Netherlands, Hungary, Qatar — highest degradation rates in the era dataset

RECOMMENDATION 02 **Compound Selection — Calibrate SOFT Stint Length Precisely by Circuit Type**

SOFT tire degradation on high-deg circuits runs at approximately 3x the rate of HARD compound degradation. This makes stint length the critical variable for any strategy involving a SOFT opening. A SOFT stint that extends past the 12–14 lap window on a high-deg circuit will erase the initial pace advantage entirely through degradation loss — the data shows the crossover point clearly in the regression slopes. On low-deg circuits the crossover extends to 18–22 laps, making longer SOFT stints viable and shifting the strategic calculation toward maximizing fresh-tire pace. Compound selection models should be circuit-type-specific and updated each season as the fleet-wide degradation rates decline with car development. A model calibrated on 2022 data underestimates tire longevity in 2024 conditions.

Most effective transition across the era: SOFT to HARD — highest association with podium outcomes and best pace-to-longevity tradeoff at low and medium degradation circuits

RECOMMENDATION 03 **Driver Coaching — Target Lap-Time Consistency in Stints 2 and 3**

Pace consistency — measured as the standard deviation of race-normalized lap times — is a meaningful predictor of finishing position that is independent of raw car speed. The ground effect era's stiff low-profile tires are more sensitive to mechanical abuse than the 13-inch tires they replaced: lateral slip, brake bias, and throttle application through high-load corners translate more directly into surface wear without the thermal buffer of a flexible sidewall. Ferrari's driver coaching program should target Stints 2 and 3 specifically, where fatigue interacts with degradation and lap-time consistency begins to decline. Reducing lap-to-lap delta in the final third of each stint has been shown across the era to extend tire life by 3–5 laps on high-deg circuits — equivalent to a free pit stop's worth of track position.

Recommended KPI: Track normalized lap-time standard deviation per stint across the season as a driver development metric alongside traditional sector time benchmarks

07 Appendix

Metrics glossary and methodology notes

Metrics Glossary

Metric	Definition
Degradation slope	Coefficient from OLS regression of lap_time_sec ~ lap_in_stint. Units: seconds per lap. Higher = faster degradation.
Lap-time normalization	Z-score within each race: (lap_time − race_mean) / race_std. Enables cross-circuit and cross-season comparison.
Strategy delta	Pace rank minus finish position per race. Positive = finished better than raw pace predicted. Negative = finished worse.
Pit window	Early = before 33% of race distance. Mid = 33–66%. Late = after 66%.
Position group	Podium = P1–P3. Top 10 = P4–P10. Back = P11 and beyond.
Compound transition	Compound removed at a stop paired with compound installed, e.g. SOFT to HARD.
High-deg circuit	Circuits where SOFT degradation exceeds ~0.08 sec/lap: Bahrain, Spain, Britain, Hungary, Netherlands, Qatar, Brazil.
Low-deg circuit	Circuits where degradation differences between compounds are small: Monaco, Monza, Singapore, Baku, Abu Dhabi, Las Vegas.

Methodology Notes

- Degradation slopes require a minimum of 5 laps per group. Groups with fewer than 5 observations return NaN and are excluded from heatmap cells.
- Strategy delta is averaged across the season per driver. Drivers with fewer than 10 race starts are excluded from era-level rankings to prevent small-sample distortion.
- The 110% threshold for safety car lap removal is consistent with FIA Article 39.1, which sets the maximum permitted lap time during yellow flag periods at 110% of the fastest qualifying time.
- Pit stop duration data is not used in this analysis — it is not consistently populated across all sessions in the FastF1 dataset for all four seasons.
- All analysis was run using Python 3.x with pandas, numpy, scipy, and matplotlib. The full pipeline is reproducible from the accompanying Python script using only the FastF1 library and publicly available F1 timing data.

Tools

Tool	Purpose	Tool	Purpose

FastF1	Data extraction from F1 API	pandas	Data wrangling and cleaning
numpy	Numerical operations	scipy.stats	Linear regression, correlations
matplotlib	Chart generation	Tableau	Interactive dashboard
Python 3.x	Full analysis pipeline	VSC	Structured data storage